

Mihir Upadhyay

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EDUCATION

New York University

M.S. in Data Science

GPA: 3.9/4.0

New York, US

Sep 2024 - Present

Indian Institute of Technology BHU

B.Tech in Chemical Engineering

GPA: 8.68/10.0

Varanasi, India

Jul 2017 - May 2021

RESEARCH & INDUSTRY EXPERIENCE

University of Cambridge

Visiting Student Researcher

New York, US

Sep 2025 - Present

- Developing algorithms to analyze and improve human-AI collaboration in multi-agent systems.

Research Center Trustworthy Data Science and Security, Ruhr University

Research Intern

Bochum, Germany

May 2025 - Aug 2025

- Systematically analyzed methods for evaluating the diversity of information presented by generative AI content retrieval agents vs. traditional web search (Under review @ WSDM 2026).
- Curated 6 real-world query datasets and collected search results via various search APIs, analyzing source coverage and temporal patterns.
- Designed a human evaluation study using Prolific to measure user preferences across Google AI Overviews, Gemini, and GPT-4 pipelines.

Flixstock

Product Manager

Gurugram, India

Oct 2022 - Sep 2023

- Designed and deployed e-commerce image generation pipelines using diffusion models, GANs, and image captioning, enabling AI models to replace human models in 20% of the catalog.
- Directed a team of 8+ engineers to develop and launch three full-stack platforms with REST API endpoints, deployed on AWS.

Standard Chartered Bank

AI/ML Analyst

Bengaluru, India

Jul 2021 - Sep 2022

- Evaluated and deployed bias, drift, and performance monitoring tools (Truera, Datarobot) for 10+ forecasting models, benchmarking against traditional ML methods.
- Led a 5+ member team to deploy tools via OpenShift and Kubernetes (Docker containers), leveraging Kubernetes operators for lifecycle management and Jenkins for CI/CD.

Data Science Intern

May 2020 - Jul 2020

- Built and tuned ensemble classifiers on a European credit card transaction dataset, improving fraud detection and reducing false positives by 5%.
- Developed Tableau dashboards for KPIs like transaction volume and response time, reducing reporting time by 4 hours/week.

TEACHING

Teaching Assistant

Courant Institute of Mathematical Sciences, NYU

New York, US

Sep 2025 - Present

- Course: Algebra, Trigonometry, and Functions

Teaching Assistant

Center for Data Science, NYU

New York, US

Jan 2025 - May 2025

- Course: Responsible Data Science

PUBLICATIONS

When Should We Orchestrate Multiple Agents?

Under review @ NeurIPS 2025

*Umang Bhatt, Sanyam Kapoor, **Mihir Upadhyay**, Katherine M. Collins, Ilia Sucholutsky, Francesco Quinzan, Adrian Weller, Andrew Gordon Wilson, Muhammad Bilal Zafar*

Characterizing Search in The Age of Generative AI

Under review @ WSDM 2026

*Elisabeth Kirsten, Jost Grosse Perdekamp, **Mihir Upadhyay**, Krishna P. Gummadi and Muhammad Bilal Zafar*

ONGOING PROJECTS

- Building a platform to evaluate generative AI agents on a custom poetry dataset derived from high school curricula across diverse regions and languages.
- Developing a framework to assess diversity in collaborative and competitive human–AI agent settings using real-world queries across domains.

AWARDS

- **SVSAES Scholarship** for MS funding (\$75K).
- **IIT BHU Merit Scholarship** for the academic years 2017-2021.

SKILLS & INTERESTS

- **Programming:** Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Hugging Face, LangChain, Transformers), SQL, Shell/Bash, HTML, JavaScript, C/C++, Prolific, MongoDB, Docker, AWS
- **Coursework:** Deep Learning, Machine Learning, Big Data, Linear Algebra & Optimization, Probability & Statistics, Responsible Data Science